MEDIUM · ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

06

6 questions · ~9 min

Q1 ADVANCED MATH

$$y = ax^2 - c$$

In the equation above, a and c are positive constants. How many times does the graph of the equation above intersect the graph of the equation $y = a + c$ in the xy-plane?

A. Zero

B. One

C. Two

D. More than two

Q2

GRID-IN ADVANCED MATH

$$7x^2 - 20x - 32 = 0$$

What is the positive solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 ADVANCED MATH

$$3xx - 4x + 5 = 0$$

What is one of the solutions to the given equation?

A. -4

B. 0

C. 3

D. 5

Q4 ADVANCED MATH

$$52x^3 + 64x^4 - 81 = 0$$

How many distinct real solutions does the given equation have?

A. Exactly two

B. Exactly three

C. Exactly five

D. Exactly seven

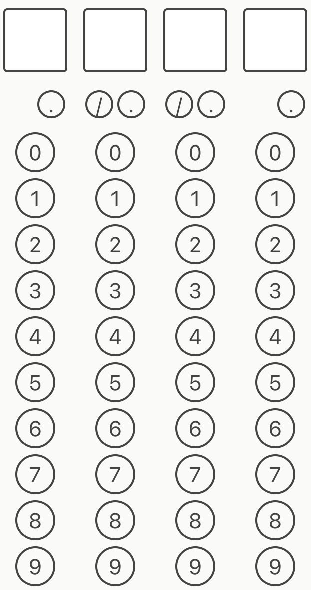
Q5

GRID-IN ADVANCED MATH

$$y = x^2 - 4x + 4$$
$$y = 4 - x$$

If the ordered pair (x, y) satisfies the system of equations above, what is one possible value of x ?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6 ADVANCED MATH

$$7m = 5n + p$$

The given equation relates the positive numbers m, n, and p. Which equation correctly gives n in terms of m and p?

A. $n = \frac{5p}{7m}$

B. $n = \frac{7m}{5} - p$

C. $n = 57m + p$

D. $n = 7m - 5 - p$